

last class:

ordered pairs: (x, y) [two things separated by a comma & $()$]

Also, we know (ordering matters)

$$(2, 4) \neq (4, 2) \quad \boxed{\text{unlike set } \{2, 4\} = \{4, 2\}}$$

some examples of ordered pairs:

$$(i) (2, 3) \quad (ii) (H, T) \quad (iii) (\{2, 5\}, \{7\}) \quad (iv) (R, (0, 0))$$

$$(v) (2, \{1, 2, 3\})$$

Now, let's define Cartesian product:

Defⁿ: The Cartesian Product of two sets $A \in B$ is a set of ordered pairs, defined as,

$$A \times B = \{ (a, b) : a \in A, b \in B \}$$

example: $A = \{k, l, m\}$ $B = \{p, q\}$

$$A \times B = \{ (k, p), (k, q), (l, p), (l, q), (m, p), (m, q) \}$$

A		<u>k</u>	<u>l</u>	<u>m</u>
B	<u>p</u>	(k, p)	(l, p)	(m, p)
	<u>q</u>	(k, q)	(l, q)	(m, q)

Also, let, $A = \{ \begin{smallmatrix} \square \\ 1 \end{smallmatrix}, \begin{smallmatrix} \square \\ 2 \end{smallmatrix}, \begin{smallmatrix} \square \\ 3 \end{smallmatrix}, \begin{smallmatrix} \square \\ 4 \end{smallmatrix}, \begin{smallmatrix} \square \\ 5 \end{smallmatrix}, \begin{smallmatrix} \square \\ 6 \end{smallmatrix} \}$

So, $A \times A$

A	1	2	3	4	5	6
A	1	(1,1) (2,1)				
2	(1,2) (2,2)					
3	(1,3) (2,3)					
4	(1,4) (2,4)					
5	(1,5) (2,5)					
6	(1,6) (2,6)					

$$6 \times 6 = 36$$

The two examples should motivate you with
Cardinality of a Cartesian Product:

$$|A \times B| = |A| \cdot |B|$$

(This theorem is for finite sets)

Problem:

$$\textcircled{1} A = \{\pi, e, o\} \text{ \& } B = \{o, i\}$$

$$(i) A \times B$$

$$(ii) \emptyset \times B = \emptyset$$

$$(iii) \{\emptyset\} \times B$$

$$\{\emptyset\} \times B = \{(a, b) \mid a \in \{\emptyset\}, b \in B\}$$

$$[A \times B = \{(a, b) \mid a \in A, b \in B\}]$$

$$\emptyset \times B = \{(a, b) \mid a \in \emptyset, b \in B\}$$

$$\textcircled{2} \{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$$

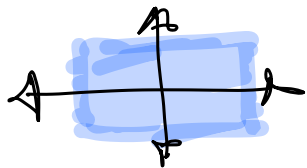
$$= \{-\sqrt{2}, \sqrt{2}\} \times \{-2, 2\}$$

$$\textcircled{3} \{x \in \mathbb{Z} : x = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$$

$$= \emptyset \times \{-2, 2\} = \emptyset$$

shifting gears to infinite set:

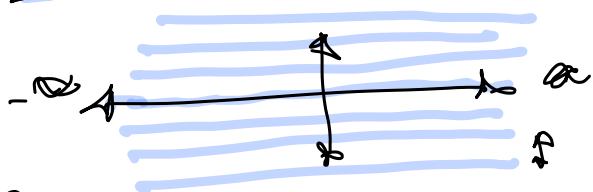
$$(i) \mathbb{R} \times \mathbb{R} = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$$



$$(ii) \mathbb{R} \times \mathbb{Z}$$

$$(iii) \mathbb{Z} \times \mathbb{R}$$

$$\bullet \mathbb{R} \times \{-3, -2, -1, 0, 1, 2, 3, \dots\}$$



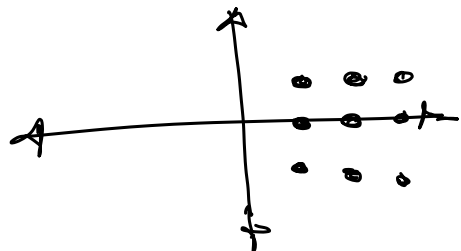
Problems:

$$\bullet \text{ sketch: } \{1, 2, 3\} \times \{-1, 0, 1\} \text{ in } \mathbb{R}^2$$

$$\bullet \text{ sketch: } \{0\} \times \mathbb{N} \text{ in } \mathbb{R}^2$$

$$(i) \{(1, -1), (1, 0), (1, 1), (2, -1), (2, 0), (2, 1),$$

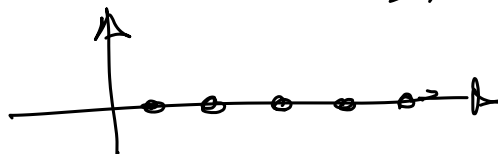
$$\dots\}$$



$$(ii) \{0\} \times \mathbb{N} = (0, \mathbb{N})$$

$$\text{in } \mathbb{R}^2, \mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$\{(0, 1), (0, 2), (0, 3), \dots\}$$



Cartesian product is not restricted to two sets

we can take Cartesian product of 3 sets, (ordered triple)

say: $A \times B \times C$, $A \times B \times C \times D$, $A \times B \times C \times \dots \times$
Ordered triple Ordered quadruple Ordered n-tuple

short note:

$$\mathbb{R} \times (\mathbb{N} \times \mathbb{Z}) \neq \mathbb{R} \times \mathbb{N} \times \mathbb{Z}$$

$(\sqrt{3}, (4, -7))$ $(\sqrt{2}, 2, -2)$

We can extend beyond 3,

$$\tilde{A}_1 \times \tilde{A}_2 \times \dots \times \tilde{A}_n = \{ (x_1, x_2, x_3, \dots, x_n) : x_1 \in A_1, x_2 \in A_2, x_3 \in A_3, \dots \}$$

where A_i are sets,

for any set A and n a positive integer

$$A^n = \underbrace{A \times A \times \dots \times A}_n = \{ (x_1, x_2, x_3, \dots, x_n) : x_1, x_2, x_3, \dots, x_n \in A \}$$

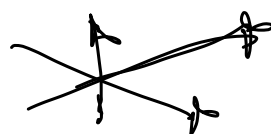
(ordered n-tuple)

examples:

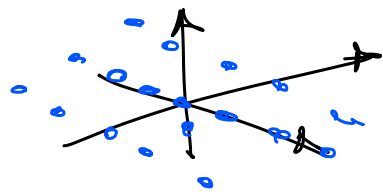
• $\mathbb{R}^2 : \{ (x, y) : x, y \in \mathbb{R} \}$



• $\mathbb{R}^3 : \{ (x, y, z) : x, y, z \in \mathbb{R} \}$



$\mathbb{Z}^3 = \{(p, q, r) : p, q, r \in \mathbb{Z}\}$
 (Cartesian power)



Problems:

(i) $B = \{0, 1\}$ find, $B \times (B \times B)$ and $\underline{B \times B \times B} = B^3$

(ii) $\{0\} \times \{0, 1\} \times \{0, 1\}$

(iii) sketch: $[0, 1] \times \{2\} \times [1, 2]$

(i) $B = \{0, 1\}$, $B \times B = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

$$\begin{aligned} B \times (B \times B) &= \{0, 1\} \times \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ &= \{(0, (0, 0)), (0, (0, 1)), (0, (1, 0)), \\ &\quad \dots \dots \dots \} \end{aligned}$$

$$\Rightarrow B \times B \times B = \{(x, y, z) : x, y, \underline{z} \in B\}$$

$$\begin{aligned} &= \{(0, 0, 0), (0, 0, 1), \\ &\quad (0, 1, 0), (0, 1, 1), \\ &\quad (1, 0, 0), (1, 0, 1), \\ &\quad (1, 1, 0), (1, 1, 1)\} \end{aligned}$$

$$\begin{aligned} B &= \{0, 1\} \\ \hat{z} &= \check{0}, \check{1} \\ y &= \check{0}, \check{1} \\ x &= \check{0}, \check{1} \end{aligned}$$

subsets:

lets start with a set,

$$B = \{0, 1, 2, 3, 4\}$$

we can construct a set by taking a subcollection from the above set, $A = \{0, 1, 2\} \rightarrow$ this is also a set

Defⁿ: suppose A and B are sets. If every element of A is also an element of B , then we say:

$$A \subseteq B$$

else,

$$A \not\subseteq B$$

In that case, A is not the subset of B

B does not contain all the elements of A

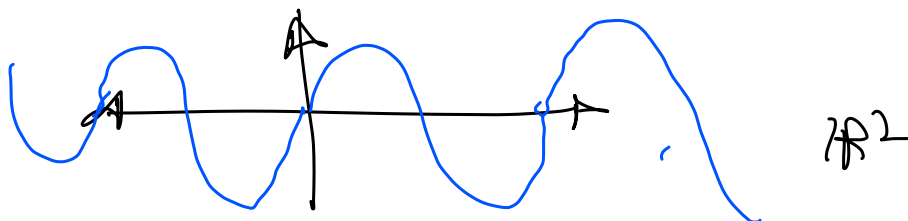
examples:

$$(i) \{2, 3, 7\} \subseteq \{1, 2, 3, 4, 5, 6, 7\}$$

$$(ii) \{2, 3, 7\} \subseteq \{2, 3, 7\}$$

$$(iii) \{(x, \sin x) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$$

graph:



$$(iv) \{3, 5, 7, 11, 13, 17, \dots\} \subseteq \mathbb{N}$$

$$(v) \{1, 3, 5, 7, \dots\} \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

$$(vi) A \subseteq A \text{ [for any set } A]$$

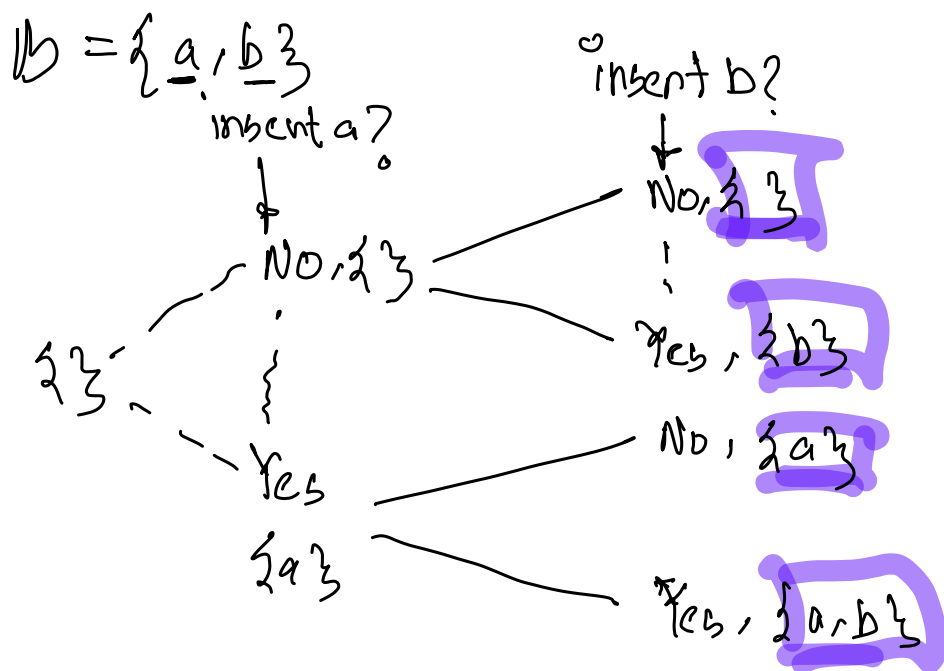
$$(vii) \emptyset \subseteq A \text{ [for any set } A], \text{ why?}$$

every element of $\emptyset$ is also an element of A

(vacuously true)

Construction of subset of a set:

using the tree diagram



$$4 = 2^2$$

no of subsets

in general: for a A with, $|A|=n$, there are 2^n no of subsets,

$$\text{for, } B = \{1, 2, \{1, 3\}\}$$

Caution:

$$\{1, 3\} \notin B, \{1, 3\} \in B$$

$$\{1\} \notin B, \{1\} \subseteq B$$

some more Caution:

(i) in $\{1, \{1\}\}$

- $\{1\} \in \{1, \{1\}\}$
- also, $\{1\} \subseteq \{1, \{1\}\}$
- and, $\{\{1\}\} \subseteq \{1, \{1\}\}$

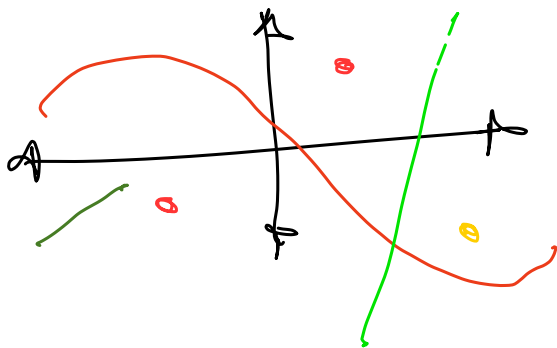
(ii) in $\{\emptyset, \mathbb{N}\}$

- $\mathbb{N} \notin \{\emptyset, \mathbb{N}\}$, $\mathbb{N} \in \{\emptyset, \mathbb{N}\}$
- $\{\mathbb{N}\} \subseteq \{\emptyset, \mathbb{N}\}$

(iii) in $\mathbb{N} \times \mathbb{N}$, = $\{(x, y) : x \in \mathbb{N}, y \in \mathbb{N}\}$

- $\{(2, 3)\} \in \mathbb{N} \times \mathbb{N}$
- $\{(2, 3)\} \subseteq \mathbb{N} \times \mathbb{N}$

subsets of $\mathbb{R}^r$:



$$\mathbb{R}^r = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$$

- collection of points
curves etc on $\mathbb{R}^r$
are subsets of $\mathbb{R}^r$

exercise:

- Determine true or false:

$$\{(x, y) \in \mathbb{R}^r : \underline{x^r - x = 0}\} \subseteq \{(x, y) \in \mathbb{R}^r : \begin{matrix} x-1=0 \\ x=1 \end{matrix}\}?$$

$$\{(0, \mathbb{R}), (1, \mathbb{R})\}^{x=0,1} \supseteq \{(1, \mathbb{R})\}$$

- $\mathbb{R}^n \leq \mathbb{R}^3$? (think about it)

Power sets:

Defⁿ: If A is a set, then the power set of A is a set which consists of, all the subsets of A

$$\mathcal{P}(A) = \{ \text{All possible subsets of } A \}$$

For a finite set A , there are $2^{|A|}$ no of subsets.

so, $|\mathcal{P}(A)| = 2^{|A|}$

example: $\mathcal{P}(\{ \{a, b\}, \{c\} \}) = \{ \emptyset, \{a, b\}, \{c\}, \{ \{a, b\}, \{c\} \} \}$

$\underbrace{\qquad\qquad\qquad}_{2^2=4}$

• Also, $\mathcal{P}(\{ \emptyset, 5 \}) = \{ \emptyset, \{ \emptyset \}, \{ 5 \}, \{ \emptyset, 5 \} \}$

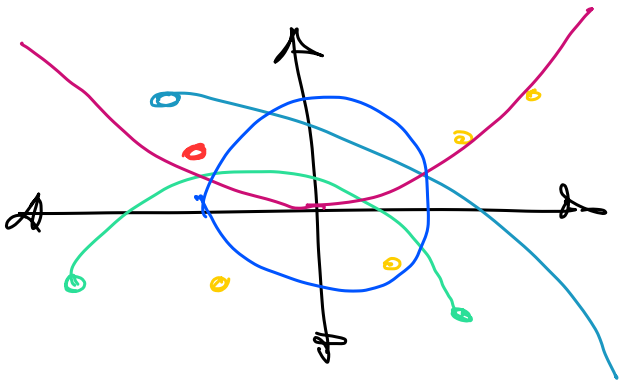
• compute: $\mathcal{P}(\{1, 2\}) \times \mathcal{P}(\{3\})$

$$= \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \} \times \{ \emptyset, \{3\} \} = \{ (\emptyset, \emptyset), (\{1\}, \emptyset), (\{2\}, \emptyset), \dots \}$$

• compute $\mathcal{P}(\{\mathbb{Z}, \mathbb{N}\})$

$$= \{\emptyset, \{\mathbb{Z}\}, \{\mathbb{N}\}, \{\mathbb{Z}, \mathbb{N}\}\}$$

Power set of $\mathbb{R}^2$:



$\mathcal{P}(\mathbb{R}^2) = \{ \text{collection of all possible subsets of } \mathbb{R}^2 \}$

$$\{x : x \in \mathcal{P}(A), |x| = 1\}$$

• for $|A|=3$, $|B|=4$

(i) Find $|\mathcal{P}(A) \times \mathcal{P}(B)| = ?$ (ii) Find $|\mathcal{P}(A \times \mathcal{P}(B))|$

(iii) $|\mathcal{P}(\mathcal{P}(A \times B))|$ (iv) $|\{x \subseteq \mathcal{P}(A) : |x| \leq 1\}|$

$$(i) \mathcal{P}(A) = 2^{|A|} \quad \mathcal{P}(B) = 2^{|B|}$$

$$|\mathcal{P}(A) \times \mathcal{P}(B)| = |\mathcal{P}(A)| \cdot |\mathcal{P}(B)| = 2^{|A|} \cdot 2^{|B|}$$

$$(ii) A \times P(B) = \underline{3 \cdot 2^4}, \quad \underline{P(A \times P(B))} = 2^{3 \cdot 2^4}$$

$$(iv) A = \{a, b, c\}, \quad \underline{P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \dots\}}$$

$$= 4$$

Union, Intersection, Difference

We start with two sets A & B ,

$$\text{union: } A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$\text{intersection: } A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$$\text{difference: } A - B = \{x : x \in A \text{ and } x \notin B\}$$

Problems:

$$A = \{a, b, c, d, e\}, \quad B = \{d, e, f\}, \quad C = \{1, 2, 3\}$$

$$\bullet A - B =$$

$$\bullet B - A =$$

$$\text{note} \\ [A \cup B = B \cup A, \\ A \cap B = B \cap A]$$

$$\bullet (A - B) \cup (B - A) =$$

$$\bullet (A \cap B) \times B =$$

some more problems:

$$A = \{0, 1\}$$

$$B = \{1, 2\}$$

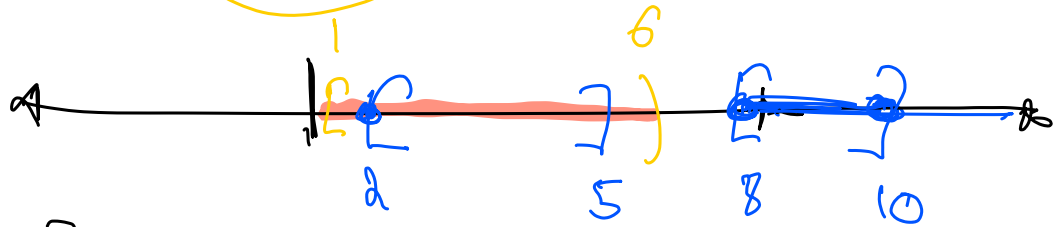
- $\mathcal{P}(A \times B)$

- $\mathcal{P}(A) \cap \mathcal{P}(B)$

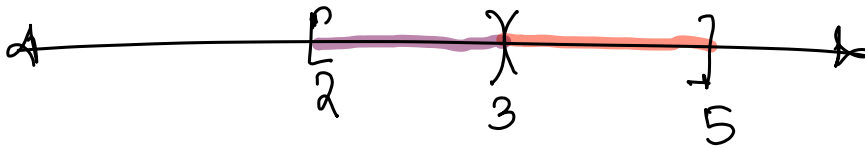
intervals:

- $([2, 5] \cup [1, 6]) - [8, 10]$

$[1, 6)$



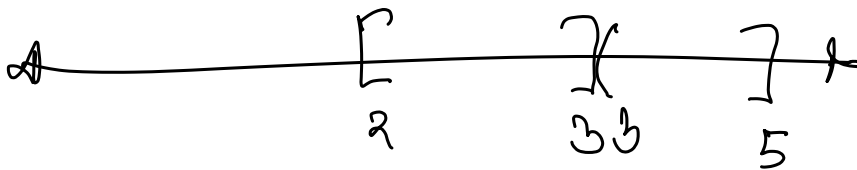
- $[2, 3) \cup (3, 5]$



do I get?

- $[2, 5]$ no!

- $[2, 3] \cap (3, 5]$



do I get
 $\{3\}$?

- $[0, 3] - [1, 2]$

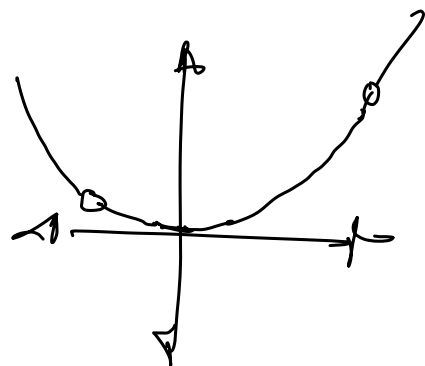
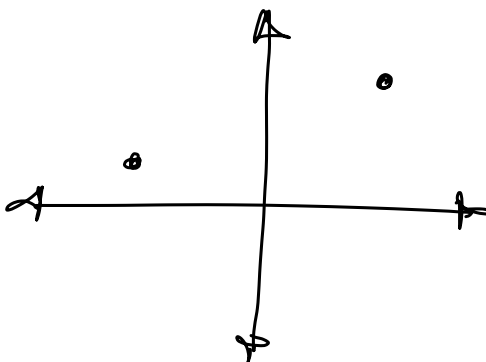
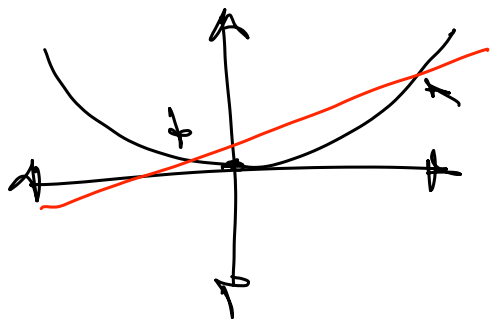
• example: $A = \{(x, x^2) : x \in \mathbb{R}\}$

Also, $B = \{(x, y) : y = x+1, x \in \mathbb{R}\}$

(i) $A \cup B$

(ii) $A \cap B$

(iii) $A - B$



• Think about:

• example: $A = \{(x, x^2) : x \in \mathbb{N}\}$

Also, $B = \{(x, y) : y = x+1, x \in \mathbb{R}\}$

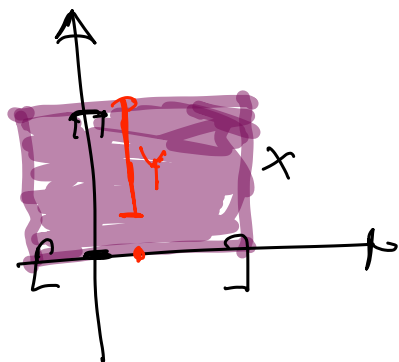
(i) $A \cup B$

(ii) $A \cap B$

(iii) $A - B$

• sketch, $X = [-1, 3] \times [0, 2]$ $Y = \{1\} \times [1, 2]$

Now locate, $X \cap Y$, $X - Y$



Compliment:

Universal set:

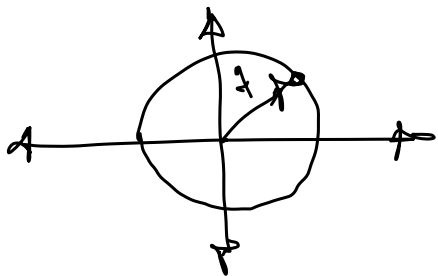
When dealing with sets, in almost all the cases, that set is a part of a larger set.

say, the set of Primes, P

$$P = \{1, 2, 3, 5, 7, 11, 13, \dots\}$$

so, Naturally it is a subset of $\mathbb{N}$

Now, for, Circle in $\mathbb{R}^2$



$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

Defn of compliment:

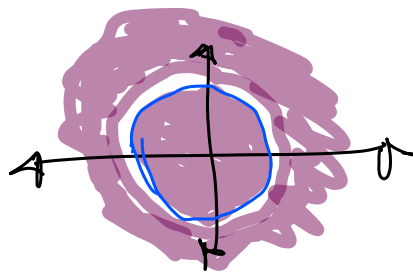
let A be a set, with U universal set.

The compliment of A ,

$$\bar{A} = U - A$$

$$\bullet \bar{P} = \mathbb{N} - P = \{4, 6, 8, 9, 10, 12, \dots\}$$

$$\bullet \bar{C} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \neq 1\}$$



Problems:

$$A = \{0, 2, 4, 6, 8\}, B = \{1, 3, 5, 7\}$$

$$U = \{0, 1, 2, 3, \dots, 8\}$$

$$(i) A \cap \bar{A}$$

$$(ii) \overline{A \cup B}$$

$$(iii) \bar{A} \cap \bar{B}$$

$$(iv) A - \bar{A}$$

$$(v) \bar{A} \times B$$

$$(vi) \overline{A \cap B}$$

$$(vii) \bar{A} \cup \bar{B}$$

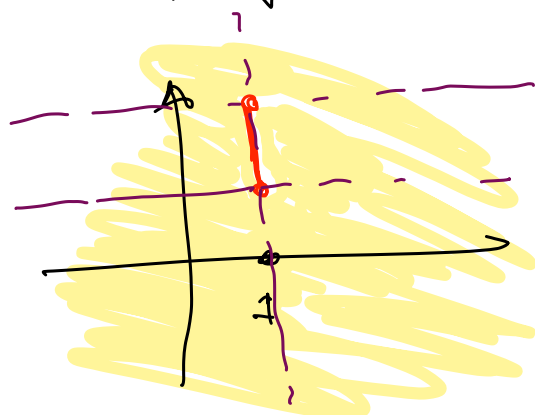
$$\bullet \text{Sketch: for } X = \{1\} \times [2, 4]$$

(i) Find, $\bar{X}$

$$(ii) \text{Also, find: } \bar{X} \cap ([0, 2] \times [0, 3])$$

$$X = \{(x, y) \in \mathbb{R}^2 : x = 1, 2 \leq y \leq 4\}$$

(i)



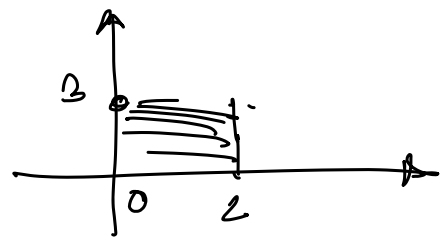
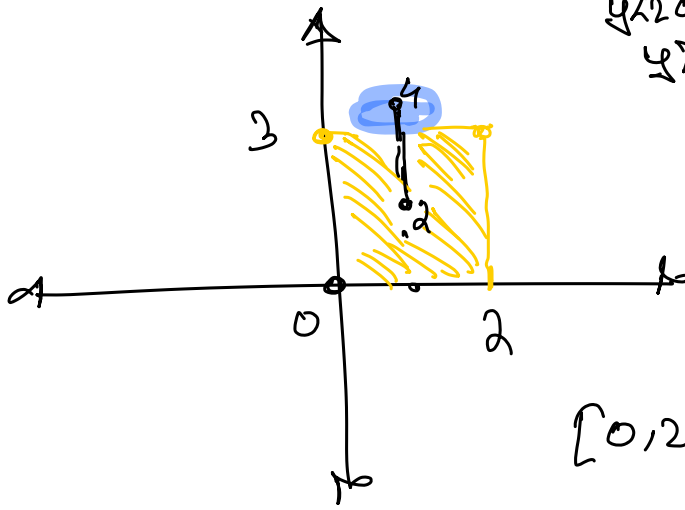
$$\bar{X} = \{(x, y) \in \mathbb{R}^2 : \text{if } x \neq 1, y \notin [2, 4]\}$$

$$\bullet \bar{X} = \{(x, y) \in \mathbb{R}^2 : (x, y) \notin \{1\} \times [2, 4]\}$$

ii)

$\bar{X} = \{ (x,y) \in \mathbb{R}^2 : x \neq 1 \text{ and}$

$y \leq 2 \text{ or } y \geq 4 \}$



$$[0,2] \times [0,3] - \{1\} \times [2,3]$$

• Sketch the set $X = \{ (x, y) \in \mathbb{R}^2 : y < x^2 \}$

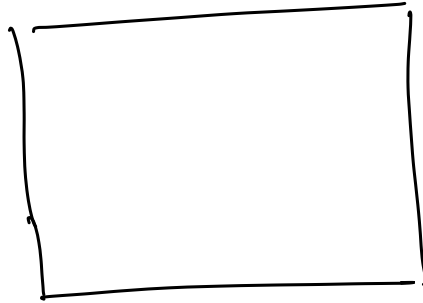
Also find, $\overline{X} \cap ([2, 3] \times [1, 2])$

Venn diagram:

Cartoonish way to represent represent set

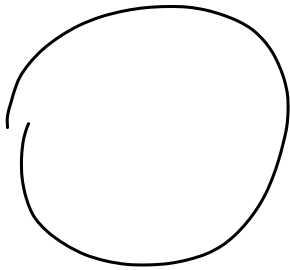
Universal set:

$U =$

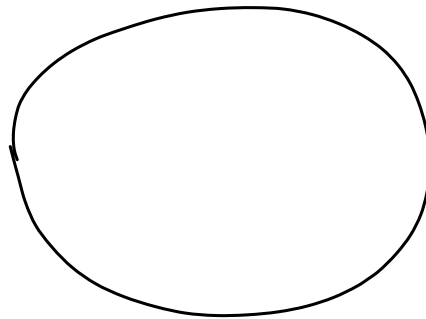


Arbitrary set:

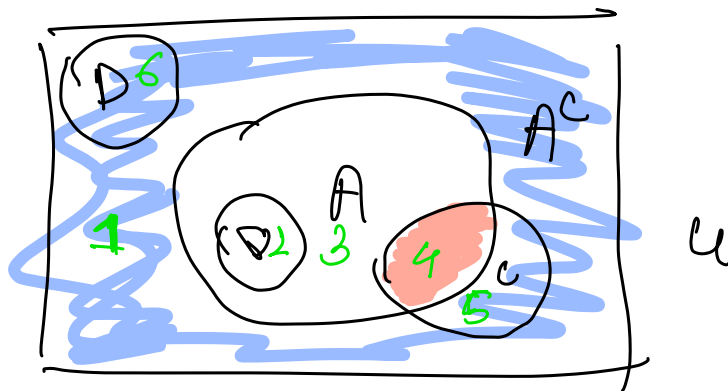
$A =$



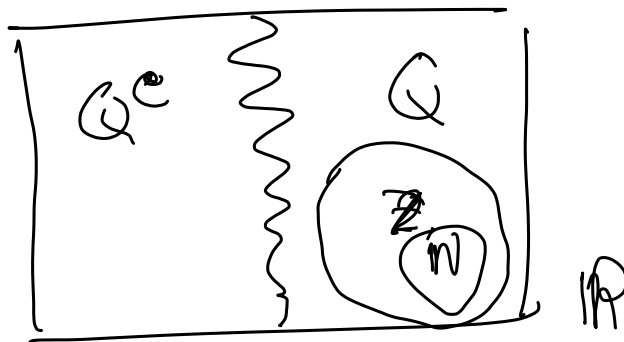
$B =$



Venn diagram



Real:



indexed set:

in maths we will be dealing with a lot of sets
or rather than writing: $A, B, C, \dots$ as sets
we can use: $A_1, A_2, A_3, \dots$

Defⁿ: suppose: $A_1, A_2, \dots, A_n$ are sets, then

$$\begin{aligned} A_1 \cup A_2 \cup A_3 \dots \cup A_n &= \{x: x \in A_1 \text{ or } x \in A_2 \text{ or } x \in A_3 \dots x \in A_n\} \\ A_1 \cap A_2 \cap A_3 \dots \cap A_n &= \{x: x \in A_1 \text{ \& } x \in A_2 \text{ \& } x \in A_3 \dots\} \end{aligned}$$

↓ (in short hand notation)

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup A_3 \cup A_4 \dots \cup A_n$$

also,

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap A_3 \dots \cap A_n$$

we can take $n \rightarrow \infty$

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \dots$$

$$\bigcap_{i=1}^{\infty} A_i =$$

ex:

$$A_1 = \{0, 2, 5\}, A_2 = \{1, 2, 5\}, A_3 = \{2, 5, 7\}$$

$$\bigcup_{i=1}^3 A_i = \{0, 1, 2, 5, 7\} = \{x: x \in A_1 \text{ or } x \in A_2 \text{ or } x \in A_3\}$$

$$\text{or, } \bigcup_{i \in I} A_i = A_1 \cup A_2 \cup A_3 \quad \{x: x \in A_i \text{ for at least one set } A_i \text{ with } i \in I\}$$

Also,

$$\bigcup_{i=1}^{\infty} A_i =$$

this can also be written as

$$\bigcup_{i \in \mathbb{N}} A_i = \{x : x \in A_i \text{ for at least one set } A_i \text{ with } i \in \mathbb{N}\}$$

we doing operation on set A_i
where i is from a set, I

I : Index set

formally, we write it as:

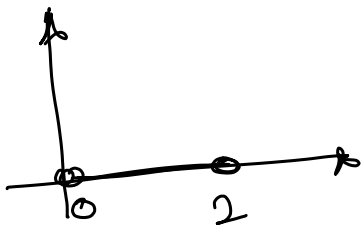
if A_α is a set for every α is in some index set:

- Index set can be any set,
 $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$, some interval

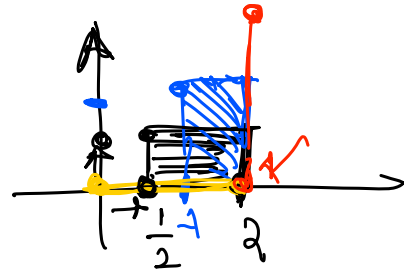
example: let, $I = [0, 2] = \{x \in \mathbb{R} : 0 \leq x \leq 2\}$

for each $\alpha \in I$, we define $A_\alpha = [\alpha, 2] \times [0, \alpha]$

$$A_{\alpha \in I} \quad A_0 = [0, 2] \times [0, 0] = [0, 2] \times \{0\}$$



$$A_{\frac{1}{2}} = [\frac{1}{2}, 2] \times [0, \frac{1}{2}]$$



• Now evaluate, $\bigcup_{\alpha \in I} A_\alpha$, $\bigcap_{\alpha \in I} A_\alpha = \{ (0, 2) \}$

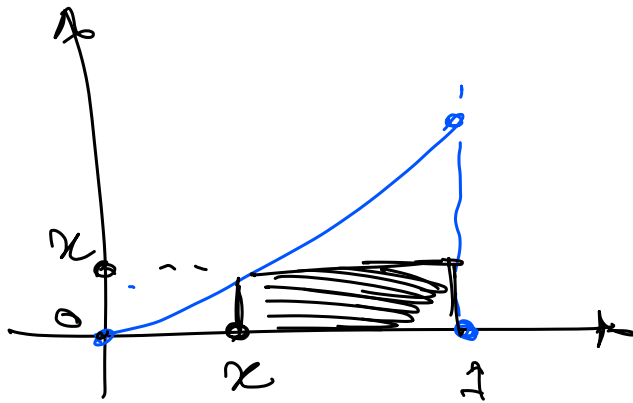
$I = [0, 2]$ right angle triangle

$\bigcup_{\alpha \in I} A_\alpha = \{x : x \in A_\alpha \text{ for at least one set } A_\alpha \text{ where } \alpha \in [0, 2]\}$

examples • $\cup$

$$[x, 1] \times [0, x^v]$$

$$x \in [0, 1]$$



$$A_x =$$

• if $J \neq \emptyset$, if $J \subseteq I$, then is:

$$(i) \bigcup_{j \in J} A_j \subseteq \bigcup_{i \in I} A_i \quad \checkmark$$

$$(ii) \bigcap_{j \in J} A_j \subseteq \bigcap_{i \in I} A_i \quad \times$$

Relation :

Cartesian product ✓

$$A \times B = \{ (x, y) : x \in A, y \in B \}$$

If we impose a condition between a and b , we get sth known as relation.

2 types :

(i) Equivalence relation: (ii) order relation

(i) Equivalence relation;

$$x, y, z \in X$$

• Reflexive: $x \sim x$

• Symmetric: $x \sim y, y \sim x$

• Transitive: $x \sim y, y \sim z \Rightarrow x \sim z$

Set: $\mathbb{N} = \{1, 2, 3, 4, 5, 6, \dots\}$

$\mathbb{N} \pmod{5}$, we are saying $x \in \mathbb{N}, y \in \mathbb{N}$
 $x \sim y$ if $x \pmod{5} = y \pmod{5}$

• How can we show this is an equivalence relation?

i) Reflexive: $x \sim x$ since, $x \pmod{5} = x \pmod{5}$

ii) Symmetric: $x \sim y \Rightarrow y \sim x$

We know, $x \sim y$ means $x \pmod{5} = y \pmod{5}$

$$\begin{array}{l} \Rightarrow y \pmod{5} = x \pmod{5} \\ \hline \downarrow \\ y \sim x \end{array}$$

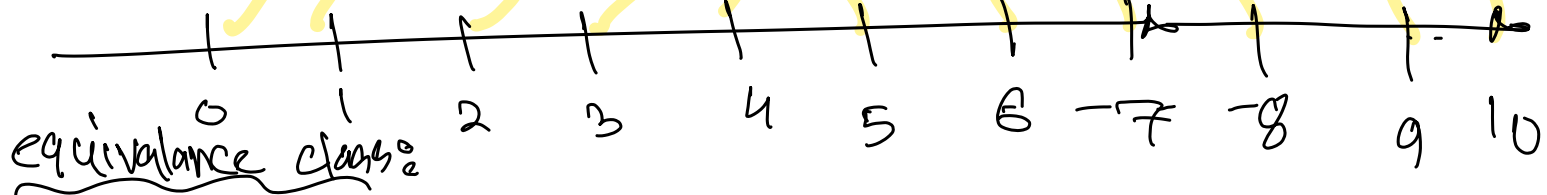
(iii) transitivity:

$$\underline{x \sim y}, \underline{y \sim z} \Rightarrow x \sim z$$

$$x \pmod{5} = y \pmod{5} \quad y \pmod{5} = z \pmod{5}$$

$$\therefore x \pmod{5} = z \pmod{5}$$

$$\therefore x \sim z$$



{ 2, 7, 12, 17, 22, ... }

{ 1, 6, 11, 16, 21, 26, 31, ... }

{ 0, 5, 10, 15, 20, 25, ... }

every equivalent class is disjoint?

example: on $\mathbb{Z}$ $a \sim b \Rightarrow |a| = |b|$

show that this is an equivalence relation:

• Reflexive: $a \sim a \Rightarrow |a| = |a|$

• Symmetric: $a \sim b \Rightarrow b \sim a$

$$\begin{array}{ccc} \downarrow & & \uparrow \\ |a| = |b| & \rightarrow & |b| = |a| \end{array}$$

• Transitivity: $a \sim b, b \sim c, a \sim c$

$$\begin{array}{ccccc} \downarrow & & \downarrow & & \nearrow \\ |a| = |b| & & |b| = |c| & \rightarrow & |a| = |c| \end{array}$$